EE2028 design vs as-built

Progress against Assignment2 enhancements, Darren's tasks, Next_Steps, and the integration / telemetry plans. Judged from repo docs + App/Core/Command (no .ioc / Drivers in this checkout). UAV / flight is out of scope (instructor).

Key finding: Software for the committed scope (#1–#4 + baseline) is largely written and documented. The hard gate is still CubeIDE + on-board verification — you cannot flash or close the mandatory acceptance checklist from this checkout alone.

~90% Baseline source	**~95%** Enhancements #1–#4	**0%** HW acceptance	**docs** Stretch #5–#6

Remaining effort (committed path, excl. stretch/UAV): ~55% source done · ~10% VESPER polish · ~25% Cube/HAL/BSP · ~10% on-board acceptance.

Workstreams

Area	Design intent	Status	Notes
Mandatory state machine	3 modes, hold-to-fire, rescue / 10s / 5 fails, Cheerio halt	Source done	Core/Src/main.c + drone.h; not flashed / acceptance unchecked
Sensors + thresholds	T/P/H/A/G/M units, any-violation pause, upside-down → Last	Call sites only	BSP paths in code; Drivers/ + I2C2 Cube config missing
Button / LED	Dual-edge EXTI, debounce, single/double/hold classifier	Logic in source	Needs Cube EXTI + pin aliases; on-board not verified
Mission UART / EVT	USART1 VCP + Telemetry_* + EVT contract	Helpers ready	Weak WriteLine → printf; HAL example not linked
#1 Matrix HUD	HT16K33 + priority display, 10s rescue clock	Implemented	Bench/hardware harness done; live EE_MATRIX_I2C pending
#2 IR threat	Observe-only chirp + log (D3/PB0)	Implemented	Bench-verified; live path needs Cube GPIO
#3 Buzzer + relay	Patterns on D2; CAP_EXPLODE on D4 at self-destruct	Mostly done	Buzzer HW-verified; relay LED/load + power-cycle check open
#4 VESPER Command	Fixture replay + Web Serial + same parser	UI ready	Live board stream blocked on USART retarget
Enhancement adapter	Thin Enhancements_* from DroneStatus + OnEvent	Merged	Behind -DEE_ENHANCEMENTS=1; integration plan goals met in source
CubeIDE / HAL / BSP	B-L4S5I-IOT01A .ioc, clocks, USART1, I2C2, Drivers/	Missing	No .ioc / Drivers in checkout — flash blocked
#5–#6 stretch	mmWave / RD-03E aim + Wi-Fi/BLE	Docs only	Deferred until mandatory HW demo; UAV out of scope (instructor)

Against the original proposals

Done (design → source)

- Shared adapter + live/bench builds (integration plan §2)
- One 10 s rescue clock feeding matrix + buzzer
- Hold-to-fire + IR observe-only (frozen rules)
- Matrix / buzzer / IR / relay modules + harnesses
- Telemetry_* + EVT helpers; VESPER fixtures + Web Serial
- Docs sync (feature overview plan — completed)

Still open (design → HW)

- Cube .ioc, HAL, BSP, peripheral handles
- USART retarget for mission stream
- Mandatory acceptance checklist (all unchecked)
- Live matrix I2C bind; relay power-cycle / load test
- Real mission UART capture → fixture
- RD-03E / mmWave + Wi-Fi stretch (UAV out of scope)

Enhancement backlog

#1 Matrix — source+ | **#2 IR** — source+ | **#3 Last/explode** — source+ | **#4 VESPER** — UI; live pending
#5 mmWave / RD-03E — not started | **#6 Wi-Fi/BLE** — not started | **UAV** — out of scope (instructor)

Immediate next five (from Next_Steps)

1. Generate & commit B-L4S5I-IOT01A Cube .ioc; link App/ + Core/
2. Prove LED + PB EXTI + USART1; enable telemetry_uart_hal
3. I2C2 / BSP WHO_AM_I for all six sensors

4. Run mandatory basic-task acceptance checklist on hardware
5. Enable EE_ENHANCEMENTS; wire D2/D3/D4 + matrix; live VESPER

Refs: docs/Next_Steps.md, Assignment2_Enhancements.md, Darren's_Tasks.md, Telemetry_EVT_Contract.md, .cursor/plans/integration.plan.md · Generated from design-progress canvas content.